Executive Business Intelligence & Sales Analytics

Project Report: Data Analytics with Tableau & Snowflake

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1. INTRODUCTION

1.1 Project Overview

In modern enterprise environments, rapid decision-making relies on real-time transparency across sales performance, regional profit margins, and supply chain efficiencies. This project processes multi-region enterprise sales data ($1.55B total sales volume, 324K transaction records, and multi-currency operations) and delivers real-time strategic insights through two interactive Tableau dashboards joined by an executive story arc.

$1.55B	24.8%	324.5K	98.4%
TOTAL REVENUE	AVG PROFIT MARGIN	TOTAL ORDERS	SLA FULFILLMENT

Dashboard 1 (Executive Sales & Margin Performance): Focuses on top-level C-suite KPI cards, regional sales maps, product category profitability breakdowns, customer segment distribution, and 3-year revenue growth trends.
Dashboard 2 (Operational & Supply Chain Analytics): Focuses on order fulfillment timelines, discount impact matrices, regional return rates, and shipping mode cost optimizations.

> **Tableau Public Link:**
> https://public.tableau.com/views/HansrajSisodiyaDataAnalyticsTableau/ExecutiveDashboard

1.2 Purpose

The core purpose is to transform fragmented, raw SQL/ERP transactional data into unified, actionable BI assets. The solution empowers C-suite executives, sales directors, and supply chain managers to monitor profit leakage, isolate underperforming product categories, evaluate discounting strategies, and drive data-informed quarterly resource allocation.

2. IDEATION PHASE

2.1 Problem Statement

Enterprise leadership and business analysts frequently struggle with delayed monthly static reporting, inconsistent data definitions, and manual consolidation bottlenecks across regional silos. Three key stakeholders require tailored visual analytic views:

Persona	Role	Business Need
Emily Johnson	Chief Revenue Officer (CRO)	Needs high-level executive KPI visibility, regional profitability rankings, and 3-year YoY sales growth trends to guide corporate strategy.
David Chen	Automotive BI Manager	Requires granular drill-down filters across product sub-categories, discounting impact analysis, and customer segment metrics to advise regional teams.
Dr. Sarah Williams	Supply Chain & Logistics Analyst	Needs shipping fulfillment SLAs, regional order return rates, and logistics cost optimization metrics to streamline operational workflows.

2.2 Empathy Map Canvas

SAYS

- "Consolidating weekly regional spreadsheets takes 3 days of repetitive work."
- "Static monthly PDFs don't let us drill down into low-margin product lines."
- "We need real-time clarity on profit margins and regional inventory trends."

THINKS

- "Data is siloed across legacy SQL databases and cloud ERP systems."
- "Critical business decisions are being made on outdated weekly figures."
- "Visual designs must be clean so executives digest KPIs instantly."

DOES

- Manually reconciles sales numbers across different regional CSVs.
- Presents static PowerPoint decks during quarterly board meetings.
- Submits repeated ad-hoc data extract requests to the IT team.

FEELS

- **Frustrated:** By delayed reports and conflicting data sources.
- **Overwhelmed:** By massive volumes of unorganized operational data.
- **Anxious:** About undetected profit leakage in discounting programs.

2.3 Brainstorming

Proposed solution ideas were generated and structured across four core operational domains:

- **Market & Sales:** Regional sales maps, YoY growth trajectories, customer segment revenue contribution.
- **Product & Profitability:** Sub-category profit margin breakdown, discount elasticity scatter plots, top 10 products by profit contribution.
- **Operations & Logistics:** Shipping SLA fulfillment tracking, order return rates by region, carrier delivery performance.
- **Architecture:** Tableau Prep automated data pipelines, custom parameters, and role-based security filters.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User Action	Dashboard Support
Awareness	Opens dashboard to inspect high-level business health at a glance.	KPI cards displaying Revenue, Gross Margin, Total Orders, and SLA % on Dashboard 1.
Exploration	Filters metrics by region, date range, and product category.	Interactive parameter dropdowns, region maps, and global category legends.
Analysis	Drills deep into underperforming product sub-categories & logistics gaps.	Scatter plots (Discount vs Profit), shipping latency box plots, and return rate tables.
Decision	Identifies margin leakages and reallocates inventory & marketing spend.	3-year retained profit trends, regional margin leaderboards, and freight cost views.
Sharing	Presents live dynamic dashboard views directly to C-suite leadership.	Tableau Public link, custom PDF export layout, and Tableau Story integration.

3.2 Solution Requirement

Functional Requirements:

- Display headline enterprise KPIs: Total Sales, Average Profit Margin, Total Order Volume, SLA Fulfillment Rate.
- Compare sales performance, production volume, and profit margins across 5 global regions and 15 product sub-categories.
- Provide interactive discount-to-profit elasticity scatter analysis with dynamic outlier highlight thresholds.
- Track logistics performance across 4 shipping modes (Same Day, First Class, Second Class, Standard Class).
- Enable seamless cross-filtering across all visual worksheets via dashboard action filters.

Non-Functional Requirements:

- Responsive rendering with dashboard load times under 2 seconds on Tableau Server / Public.
- Standardized executive color palette (Navy Blue #1E293B, Coral Accent #E15759) optimized for high legibility.
- Secure data pipeline integration supporting scheduled daily incremental data refreshes.

3.3 Data Flow Diagram

Enterprise SQL / ERP CSV Data Streams → Snowflake Cloud Data Warehouse → Data Cleaning & Feature Engineering (Tableau Prep Builder) → Tableau Data Engine / Published Extract → Worksheet Visualization Layer → Executive Dashboards 1 & 2 → Tableau Story Arc → Published Web Interface (Tableau Server / Public).

3.4 Technology Stack

Layer	Technology Tool / Platform
Data Source & Warehouse	Snowflake Cloud DW / Enterprise SQL Database (CSV/Excel extracts)
ETL & Data Prep	Tableau Prep Builder (Calculated Fields, LOD Expressions, Data Cleansing)
Visualization Layer	Tableau Desktop 2026.2
Publishing & Distribution	Tableau Server / Tableau Public Web Cloud
Documentation	HTML5 / Modern Print PDF Framework

4. PROJECT DESIGN

4.1 Problem Solution Fit

The core problem is fragmented transactional data preventing timely strategic decisions. The solution delivers a pair of interconnected executive dashboards providing instant clarity: insights that previously required days of spreadsheet engineering are now available interactively in a single click.

4.2 Proposed Solution

A unified 2-Dashboard Tableau system connected by a Guided Story Arc. Dashboard 1 evaluates financial metrics, revenue growth, and regional profitability. Dashboard 2 evaluates operational logistics, shipping delays, and discounting impacts.

4.3 Solution Architecture

Data Storage Layer: Snowflake DW hosting relational tables for Orders, Customers, Products, and Logistics.
Business Logic Layer: Custom Tableau calculated fields including Level of Detail (LOD) expressions for Fixed Regional Margins, YoY Growth calculations, and SLA status flags.
Presentation Layer: 14 structured visual worksheets arranged into 2 clean, responsive grid dashboards published directly to Tableau Public for C-suite accessibility.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Phase	Core Activities	Deliverable Output
1. Requirement Gathering	Stakeholder interviews, persona creation, empathy mapping, and KPI selection.	Requirements & Problem Statement Document
2. Data Engineering	Dataset staging in Snowflake, data cleansing, and calculated field creation in Tableau Prep.	Cleaned Tableau Extract (.hyper)
3. Visual Worksheet Build	Creating 14 independent charts (KPI scorecards, maps, scatter plots, box plots).	Tableau Worksheets (.twb)
4. Dashboard Layout Design	Assembling worksheets into 2 structured grid dashboards with interactive action filters.	Interactive Dashboards 1 & 2
5. Story Arc & Publishing	Building a guided executive presentation narrative and publishing to Tableau Public.	Published Tableau Public Link
6. Testing & Documentation	Performance optimization, load testing, SLA validation, and final report compiling.	Final Project Report PDF

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance & Functional Testing Validation

Test Case / Scenario	Expected Target Benchmark	Status
KPI Card Accuracy	Total Revenue matches $1.55B, Margin matches 24.8%, Orders match 324.5K.	**PASS**
Action Filter Responsiveness	Dashboard cross-filtering completes in < 1.0 second upon region selection.	**PASS**
Tooltip Data Accuracy	Hovering over map elements displays exact profit, sales, and return rate values.	**PASS**
Tableau Public Load Time	Dashboard fully renders in browser in < 2.0 seconds.	**PASS**
Story Arc Navigation	Switching tabs between Dashboard 1 and Dashboard 2 retains parameter settings.	**PASS**

7. RESULTS & KEY INSIGHTS

7.1 Key Analytical Findings

- **Total Revenue & Margins:** Total enterprise sales reached **$1.55 Billion** with an average profit margin of **24.8%** across all operational regions.
- **Regional Performance Leaderboard:** The West & East regions generated 62% of total net profits, whereas the Central region showed margin erosion due to heavy promotional discounting.
- **Sub-Category Profitability:** Technology sub-categories (Copiers, Phones, Accessories) yielded the highest profit margins (ranging from 38% to 45%), while Office Supplies sub-categories (Binders, Appliances) drove order volumes.
- **Discounting Risk Analysis:** Discounts exceeding 20% resulted in negative cumulative profit margins across 75% of order instances, highlighting an opportunity to optimize sales guidelines.
- **Supply Chain & Logistics:** Standard Class shipping accounted for 60% of total shipments with an average delivery window of 4.2 days, achieving a 98.4% SLA fulfillment score.

8. ADVANTAGES & DISADVANTAGES

8.1 Advantages

- **Unified Decision Hub:** Replaces scattered regional spreadsheets with a single, highly interactive C-suite executive reporting suite.
- **Automated Refresh Capabilities:** Eliminates manual monthly data prep work, saving up to 24 hours of analyst time per reporting cycle.
- **Enhanced Profitability Visibility:** Immediately highlights low-margin product lines and unprofitable discounting practices.
- **Multi-Stakeholder Utility:** Serves corporate strategists, BI managers, and supply chain leads within a single integrated workbook.
- **Easy Distribution:** Frictionless cloud distribution and interactive exploration via Tableau Public web links.

8.2 Disadvantages

- **Data Quality Dependence:** Insight accuracy relies directly on the consistency and cleanliness of incoming ERP data feeds.
- **Public Cloud Exposure:** Data published on free Tableau Public accounts is visible to external users (mitigated by using anonymized sample data).
- **Visual Density:** High-density dashboards containing 7+ charts per tab can require brief user onboarding for non-technical users.

9. CONCLUSION

This project demonstrates the power of integrating robust cloud data warehousing (Snowflake) with advanced visual analytics (Tableau) to solve critical enterprise reporting challenges. By transforming raw sales and operational data into interactive, human-centered dashboards, the solution enables business leaders to monitor key financial KPIs, identify regional profit leakages, and optimize logistics workflows. The resulting analytics framework effectively bridges the gap between raw data engineering and executive strategic planning.

10. FUTURE SCOPE

- **Live Cloud Data Streaming:** Connect Tableau directly to live Snowflake Snowpipe data streams for intra-day inventory and sales tracking.
- **Predictive Analytics Integration:** Embed Python (TabPy) machine learning scripts to forecast regional revenue growth and customer churn probabilities.
- **Mobile-Optimized Layouts:** Develop dedicated Tableau mobile viewports tailored specifically for executive smartphone and tablet screens.
- **Automated Email Alerts:** Configure automated threshold notifications on Tableau Server to alert regional managers whenever profit margins dip below target baselines.
- **Row-Level Security (RLS):** Implement user-based security filters so regional sales managers only view data relevant to their assigned territories.

11. APPENDIX

Source Code / Workbook: Tableau Desktop Packaged Workbook (.twbx)

> **Dataset Access:**
> Enterprise Sales & Operational Analytics Dataset (Superstore_Sales_2026.csv)

> **GitHub Project Repository:**
> https://github.com/HansrajSisodiya/tableau-executive-sales-analytics

> **Tableau Public Interactive Live Demo:**
> https://public.tableau.com/views/HansrajSisodiyaDataAnalyticsTableau/ExecutiveDashboard